LABSHEET-8

Q1. Draft a project specification and wireframe layout for a fictional client (e.g., local artisan bakery, eco-friendly cafe, or boutique fitness studio).

Project Name: Green Leaf Eco Cafe

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Business Type: Eco-Friendly Cafe

Target Audience:
Students, families and working professionals.

Website Pages:
1. Home
2. About Us
3. Menu
4. Our Team
5. Testimonials
6. Contact Us
7. Booking

Main Features:
1. Responsive design
2. Online table booking
3. Food gallery
4. Customer reviews
5. Google Maps
6. Social media links

Wireframe Layout:

HEADER
Logo | Home | About | Menu

HERO SECTION
Welcome to Green Leaf
Book Table

ABOUT US

OUR MENU

OUR TEAM

TESTIMONIALS

BOOKING FORM

CONTACT US

FOOTER

Q2. Implement a semantic HTML5 structure reflecting the client's business offerings, team, testimonials, and contact details.

Commands:

```
mkdir GreenLeafCafe
cd GreenLeafCafe
touch index.html style.css script.js
mkdir images
```

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Green Leaf Eco Cafe</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>

<header>
   <h1>Green Leaf Eco Cafe</h1>
   <nav>
      <a href="#home">Home</a>
      <a href="#about">About</a>
      <a href="#menu">Menu</a>
      <a href="#team">Team</a>
      <a href="#contact">Contact</a>
   </nav>
</header>

<main>
   <section id="home">
      <h2>Welcome to Green Leaf Cafe</h2>
      <p>Fresh Food, Healthy Life.</p>
      <button onclick="showMessage()">Explore More</button>
   </section>

   <section id="about">
      <h2>About Us</h2>
      <p>We serve fresh and organic food.</p>
   </section>

   <section id="menu">
      <h2>Our Menu</h2>
      <p>Organic Coffee, Fresh Salad and Healthy Meals.</p>
   </section>

   <section id="team">
      <h2>Our Team</h2>
      <p>Meet our professional cafe team.</p>
   </section>

   <section id="testimonials">
      <h2>Customer Reviews</h2>
      <p>Excellent food and friendly service!</p>
```

```
    </section>

    <section id="contact">
        <h2>Contact Us</h2>
        <p>Email: info@greenleafcafe.com</p>
        <p>Phone: 9876543210</p>
    </section>
</main>

<footer>
    <p>&copy; 2026 Green Leaf Eco Cafe</p>
</footer>

<script src="script.js"></script>
</body>
</html>
```

Q3. Develop a fully responsive CSS layout using CSS Grid and Flexbox adhering to mobile-first design principles.

style.css

```
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: Arial, sans-serif;
    line-height: 1.6;
}

header {
    background: #245c3a;
    color: white;
    padding: 20px;
    display: flex;
    flex-direction: column;
    align-items: center;
}

nav {
    display: flex;
    flex-wrap: wrap;
    gap: 15px;
}

nav a {
    color: white;
    text-decoration: none;
}

section {
    padding: 40px 20px;
    text-align: center;
}

#menu, #team, #testimonials {
```

```
    display: grid;
    grid-template-columns: 1fr;
    gap: 20px;
}

footer {
    background: #245c3a;
    color: white;
    text-align: center;
    padding: 20px;
}

@media (min-width: 768px) {
    header {
        flex-direction: row;
        justify-content: space-between;
    }

    #menu, #team, #testimonials {
        grid-template-columns: repeat(3, 1fr);
    }
}
```

Q4. Incorporate modern web design principles including consistent color palettes, typography scales, and white space management.

style.css

```
:root {
    --primary: #245c3a;
    --secondary: #a8d5a2;
    --background: #f5f8f2;
    --text: #222222;
}

body {
    background-color: var(--background);
    color: var(--text);
    font-family: Arial, sans-serif;
}

h1 {
    font-size: 2.5rem;
}

h2 {
    font-size: 2rem;
    color: var(--primary);
}

p {
    font-size: 1rem;
    line-height: 1.8;
    margin: 15px 0;
}

section {
    padding: 50px 25px;
    margin-bottom: 20px;
```

```
}
```

Q5. Implement web accessibility standards (WCAG) including ARIA landmarks, alt text for all images, and sufficient color contrast ratios.

index.html

```
<header role="banner">
   <nav role="navigation" aria-label="Main navigation">
      <a href="#home">Home</a>
      <a href="#menu">Menu</a>
      <a href="#contact">Contact</a>
   </nav>
</header>

<main id="main-content" role="main">
   <section id="home" aria-labelledby="home-heading">
      <h2 id="home-heading">Welcome to Green Leaf Cafe</h2>

      <img src="images/cafe.jpg"
          alt="Green Leaf Cafe interior">
   </section>
</main>

<footer role="contentinfo">
   <p>Green Leaf Eco Cafe</p>
</footer>
```

style.css

```
body {
   color: #222222;
   background: #ffffff;
}

a:focus, button:focus, input:focus {
   outline: 3px solid #245c3a;
   outline-offset: 3px;
}
```

Q6. Create an interactive hero carousel showcasing client products and services using JavaScript and CSS transitions.

index.html

```
<div class="carousel">
   <div class="slide active">Fresh Organic Food</div>
   <div class="slide">Healthy Drinks</div>
   <div class="slide">Natural Ingredients</div>

   <button onclick="nextSlide()">Next</button>
</div>
```

style.css

```
.slide {
```

```
    display: none;
    padding: 50px;
    background: #a8d5a2;
    text-align: center;
    transition: opacity 0.5s;
}

.slide.active {
    display: block;
}
```

script.js

```
let currentSlide = 0;
const slides = document.querySelectorAll(".slide");

function nextSlide() {
    slides[currentSlide].classList.remove("active");

    currentSlide = (currentSlide + 1) % slides.length;

    slides[currentSlide].classList.add("active");
}
```

Q7. Design and integrate an interactive booking or reservation form with client-side JavaScript validation.

index.html

```
<section id="booking">
    <h2>Book Your Table</h2>

    <form id="bookingForm">
        <input type="text" id="name"
            placeholder="Your Name" required>

        <input type="email" id="email"
            placeholder="Your Email" required>

        <input type="date" id="date" required>

        <input type="number" id="guests"
            placeholder="Number of Guests"
            min="1" max="10" required>

        <button type="submit">Book Now</button>
    </form>

    <p id="message"></p>
</section>
```

script.js

```
document.getElementById("bookingForm")
.addEventListener("submit", function(event) {

    event.preventDefault();
```

```
    let name = document.getElementById("name").value;
    let email = document.getElementById("email").value;
    let guests = document.getElementById("guests").value;

    if (name === "" || email === "" || guests < 1 || guests > 10) {
        alert("Please enter valid details");
    } else {
        document.getElementById("message").innerText =
            "Booking request submitted successfully!";
    }
});
```

Q8. Implement Google Maps embedding and responsive styling on the client's contact page.

index.html

```
<section id="contact">
    <h2>Find Us</h2>

    <iframe
        src="https://www.google.com/maps?q=Roorkee&output=embed"
        width="100%"
        height="350"
        style="border:0;"
        loading="lazy"
        title="Google Map showing Roorkee">
    </iframe>
</section>
```

style.css

```
iframe {
    width: 100%;
    max-width: 100%;
    border-radius: 10px;
}
```

Q9. Add social media integration feeds and sharing buttons within the website footer.

index.html

```
<footer>
    <h3>Follow Us</h3>

    <a href="https://www.facebook.com/">Facebook</a>
    <a href="https://www.instagram.com/">Instagram</a>
    <a href="https://www.linkedin.com/">LinkedIn</a>

    <button onclick="shareWebsite()">Share Website</button>
</footer>
```

script.js

```
function shareWebsite() {
   if (navigator.share) {
      navigator.share({
         title: "Green Leaf Eco Cafe",
         text: "Visit Green Leaf Eco Cafe",
         url: window.location.href
      });
   } else {
      alert("Sharing is not supported in this browser.");
   }
}
```

Q10. Optimize website assets (compressing images, minifying CSS/JS) for high performance and fast loading speeds.

Commands

```
npm init -y
npm install --save-dev terser clean-css-cli
npx terser script.js -o script.min.js
npx cleancss -o style.min.css style.css
```

index.html

```
<link rel="stylesheet" href="style.min.css">
<script src="script.min.js" defer></script>
```

Q11. Test website cross-browser compatibility across Chrome, Firefox, Edge, and Safari.

Browser Testing Checklist:

1. Google Chrome - Open website and test all links.
2. Mozilla Firefox - Check layout and forms.
3. Microsoft Edge - Test navigation and buttons.
4. Apple Safari - Check fonts and responsive layout.

Testing Points:
- Page loading
- Navigation links
- Images
- Booking form
- Carousel
- Footer
- CSS layout

Q12. Verify mobile responsiveness across diverse device viewports using browser developer tools.

Responsive Testing:

1. Open website in Google Chrome.
2. Press F12.
3. Click Toggle Device Toolbar.
4. Select different device sizes.
5. Check website layout.

Test Viewports:

Mobile: 375px
Tablet: 768px
Laptop: 1024px
Desktop: 1440px

Q13. Prepare a comprehensive design rationale document explaining color choices, typography, and target audience alignment.

DESIGN RATIONALE

Project Name: Green Leaf Eco Cafe

1. Color Choices:
Green represents nature and freshness.
White represents cleanliness.
Light green creates a peaceful appearance.

2. Typography:
Arial font is used for readability.
Headings are larger than paragraph text.

3. Target Audience:
Students, families and working professionals.

4. Layout:
Simple navigation and organized sections.

5. Responsive Design:
Website works on mobile, tablet and desktop.

6. Accessibility:
Alt text, readable fonts and keyboard navigation
are included.

Q14. Create a slide presentation outlining the website planning, wireframing, development phases, and testing results.

PRESENTATION: GREEN LEAF ECO CAFE

Slide 1: Project Introduction
Green Leaf Eco Cafe Website

Slide 2: Project Objectives
Online presence and customer engagement.

Slide 3: Target Audience
Students, families and professionals.

Slide 4: Wireframe Design
Website page layout.

Slide 5: Technologies Used
HTML5, CSS3 and JavaScript.

Slide 6: Development

Responsive layout and interactive features.

Slide 7: Testing
Browser and mobile testing.

Slide 8: Conclusion
Successful website development.

Q15. Deliver a live demonstration of the website functionality, interactive features, and responsive design during the lab presentation.

DEMONSTRATION STEPS:

1. Open the website in a browser.
2. Show the home page.
3. Demonstrate navigation links.
4. Show the hero carousel.
5. Fill and submit the booking form.
6. Demonstrate Google Maps.
7. Show customer testimonials.
8. Test the website on mobile view.
9. Demonstrate social sharing.

Q16. Implement SEO meta tags (title, description, Open Graph tags) across all pages of the client website.

index.html

```
<head>
   <title>Green Leaf Eco Cafe | Healthy Food</title>

   <meta name="description"
       content="Visit Green Leaf Eco Cafe for fresh organic food and healthy drinks.">

   <meta name="keywords"
       content="Eco Cafe, Organic Food, Healthy Drinks">

   <meta property="og:title"
       content="Green Leaf Eco Cafe">

   <meta property="og:description"
       content="Fresh Food, Healthy Life.">

   <meta property="og:type"
       content="website">

   <meta property="og:image"
       content="images/cafe.jpg">

   <meta name="viewport"
       content="width=device-width, initial-scale=1.0">
</head>
```

Q17. Set up a local test environment to host and run the client website smoothly before presentation.

Commands

```
cd GreenLeafCafe
php -S localhost:8000
```

Browser URL

http://localhost:8000

Q18. Incorporate customer testimonial grids with interactive rating stars.

index.html

```
<section id="testimonials">
   <h2>Customer Reviews</h2>

   <div class="review">
      <h3>Customer 1</h3>
      <p>Excellent food and service!</p>
      <div class="stars" onclick="rateUs(event)">
         ★★★★★
      </div>
   </div>

   <div class="review">
      <h3>Customer 2</h3>
      <p>Fresh and healthy food.</p>
      <div class="stars" onclick="rateUs(event)">
         ★★★★★
      </div>
   </div>
</section>
```

style.css

```
#testimonials {
   display: grid;
   grid-template-columns: 1fr;
   gap: 20px;
}

.review {
   padding: 20px;
   background: #e8f5e9;
   border-radius: 10px;
}

.stars {
   color: orange;
   font-size: 25px;
   cursor: pointer;
}

@media (min-width: 768px) {
   #testimonials {
      grid-template-columns: repeat(2, 1fr);
   }
```

```
}
```

script.js

```
function rateUs(event) {
   let stars = event.currentTarget;
   let count = Math.ceil(
      (event.clientX - stars.getBoundingClientRect().left) /
      stars.getBoundingClientRect().width * 5
   );

   count = Math.max(1, Math.min(5, count));

   stars.innerText = "★".repeat(count) + "☆".repeat(5 - count);
}
```

Q19. Write clean, well-commented HTML, CSS, and JavaScript code following industry coding standards.

index.html

```
<!-- Main Website Header -->
<header>
   <h1>Green Leaf Eco Cafe</h1>
</header>

<!-- Main Content -->
<main>
   <section id="about">
      <h2>About Us</h2>
      <p>We provide fresh and organic food.</p>
   </section>
</main>

<!-- Website Footer -->
<footer>
   <p>Copyright 2026 Green Leaf Eco Cafe</p>
</footer>
```

style.css

```
/* Main Body Styling */
body {
   font-family: Arial, sans-serif;
   margin: 0;
   padding: 0;
}

/* Header Styling */
header {
   background: #245c3a;
   color: white;
   padding: 20px;
   text-align: center;
}

/* Main Section Styling */
```

```
main {
   padding: 20px;
}
```

script.js

```
// Display welcome message
function showMessage() {
   alert("Welcome to Green Leaf Eco Cafe!");
}
```

Q20. Conduct a final code audit and accessibility evaluation using automated tools (e.g., Lighthouse).

FINAL AUDIT CHECKLIST

1. Check HTML structure.
2. Check CSS errors.
3. Check JavaScript errors.
4. Verify all images have alt text.
5. Check keyboard navigation.
6. Check color contrast.
7. Verify mobile responsiveness.
8. Test all website links.
9. Check page loading speed.
10. Run Google Lighthouse.

LIGHTHOUSE STEPS:

1. Open website in Google Chrome.
2. Press F12.
3. Select Lighthouse tab.
4. Select Performance.
5. Select Accessibility.
6. Select Best Practices.
7. Select SEO.
8. Click Analyze Page.
9. Review the generated report.